

Victoria Castor Villegas

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EDUCATION

PhD in Chemistry

University of Rouen Normandy

Oct 2022 – Oct 2025

Rouen, France

- Thesis: *Chemical Reactivity in Solution from a Hybrid Conceptual DFT and QTAIM Approach*

MSc Theoretical Chemistry and Computational Modelling

Autonomous University of Madrid

Sep 2020 – Jul 2022

Madrid, Spain

- Thesis: *Hydrogen Bonds in Water Clusters from an ELF Perspective*

BSc Chemistry

National Autonomous University of Mexico

Aug 2016 – Jul 2020

Mexico City, Mexico

- Thesis: *Topological Studies of the Electronic Density in Water Clusters*

WORK

Postdoc

University of Rouen Normandy

Dec 2025 – Present

Rouen, France

- Research: *Quantum-Chemical Modelling of NMR and Reactivity from Molecular Dynamics and Topological Analysis*

PUBLICATIONS

1. V. Castor-Villegas, José Manuel Guevara-Vela, Wilmer E. Vallejo Narváez, Ángel Martín Pendás, Tomás Rocha-Rinza, and Alberto Fernández-Alarcón. On the strength of hydrogen bonding within water clusters on the coordination limit. *Journal of Computational Chemistry*, 2020.
2. V. Castor-Villegas, Vincent Tognetti, and Laurent Joubert. On the prediction by density functional theory of entropies in solution within implicit solvation models. *Journal of Molecular Modeling* 31(1), 2024.

SKILLS

Computational Chemistry : Amsterdam Modeling Suite (ADF, BAND, DFTB), Gaussian, AIMAll, Orca, LAMMPS

Programming & Software : UNIX/Linux, Git/Subversion, Fortran, $\LaTeX$, Python (sklearn & tensorflow), SQL, C, C++, Perl, Cobol, Julia, JavaScript, WebAssembly

Visualisation & Graphics : Inkscape, GIMP, Tcl/Tk, Manim

Other : Astrophysics, First Aid

LANGUAGES

Spanish : C2 **English** : C2 **French** : B2 **German** : A2 **Chinese** : HSK 1 **Hebrew** : נ